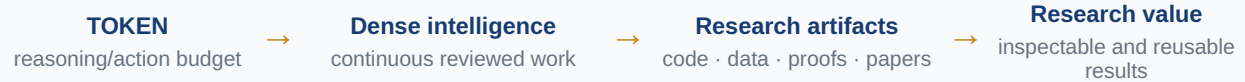
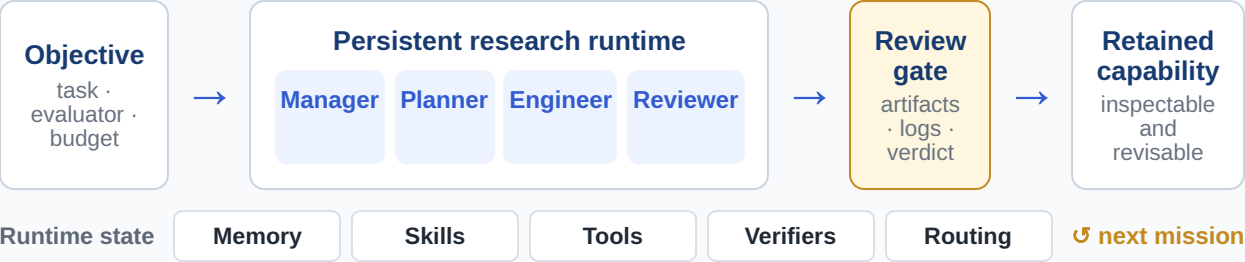


Dense-intelligence research

Continuous proposal—execution—verification over durable state

Invention Long horizon Verifiable

A Runtime architecture and fixed-model evolution



Mission trajectory π **Plan** → **Retrieve** → **Build** → **Experiment** → **Review** → **Distill** bounded outcome · persisted results

$$\rho_I(T) = T^{-1} \int \dot{N}_{\text{tok}} \eta_r \eta_a \eta_v dt$$

useful reasoning, action, and verification per unit time

$$H_{t+1} = U(H_t, \tau_t, E_t), \theta \text{ fixed}$$

accepted results update the runtime without online parameter training

B Benchmark results in task-native units

SOL-ExecBench GPT-5.5 · Codex

Global #6 · 2×
#1 · 7 top-3

B200 · 101 kernels
Two head-to-head wins over Recursive

nanochat · B200 GPT-5.5 · Codex

0.9636 BPB

5 min · 1×B200 · 426 attempts
Human SOTA: 0.9646

nanochat · H100 GPT-5.5 · Codex

0.9855 BPB

5 min · 1×H100 · 37 mechanisms
Human SOTA: 0.9879

nanoGPT speedrun GPT-5.5 · Codex

79.77 seconds

8×H100 · N=10
Same-device human #83: 80.18s

AARRI-Bench GPT-5.5 · Codex

63/82 · 76.8%

82 research-intern tasks
Paper-reported best: 68.3%

Math-Reasoning Data GPT-5.5 · Codex

28.0 gap

Math-Reasoning Data

41 de-duplicated research artifacts across 6 programs 35 manuscripts · 6 drafts

Architecture is schematic; result cards retain task-native units and are not cross-normalized.

All six runs: GPT-5.5 · Codex